

→ Density-based algorithm that discovers clusters of arbitrary shape and automatically flags outliers as noise. Perfect for anomaly detection in manufacturing process data where defects are rare but critical.

[illegible]

Silhouette Score : 0.9742

N Clusters : 3

N Noise Points : 0

AI MODEL INSIGHT

After training and comparing all recommended models, KMeans achieved the best performance. It formed 3 coherent clusters with a Silhouette Score of 0.9742 (scores near +1 indicate well-separated, compact clusters). | Model selection criterion: weighted F1 (classification) or RMSE (regression). All models trained with default hyperparameters; recommend GridSearchCV / Optuna for production hyperparameter optimisation.

■■ FEATURE INSIGHT

Feature importance scores are not directly extractable for this model type. Consider using SHAP for model-agnostic explanations. | Importances sourced from estimator's native feature_importances_ (tree-based) or mean(|coef|) (linear). For model-agnostic explanations, integrate SHAP TreeExplainer / KernelExplainer.

BIG DATA TELEMETRY & SCALE

Throughput Speed : 48.5 MB/s (595,552 records)

PySpark RAM Req. : 512.0 MB (2 executors)

Partition Column : 'author' (12 partitions)

Storage Format : Delta Lake / Apache Parquet (ACID compliant)

[illegible]

Compliance Rating: 75 / 100 (ISO 27001 Aligned)

EU AI Act Risk : Minimal Risk (Unsupervised Data Exploratory Analytics)

GDPR Privacy : Clean – No raw PII stored

Human Oversight : Mandatory (Human operator must approve critical fault actions)

→ Voluntary code of conduct and standard documentation.

ACTIONABLE RECOMMENDATIONS

1. Investigate extreme sensor spikes detected in columns ['n_models', 'total_downloads_30d'] via IQR bounds.
2. Evaluate cluster cohesion (Silhouette Score: 0.9742) using PCA 2D projections.
3. Partition Delta Lake tables on column 'author' for sub-second distributed telemetry queries.
4. Deploy the saved model binary via the FastAPI /orchestrate endpoint for production inference.

=====

